noc22_cs107

Mock Exam 4 | Course ID: noc22_cs107

Q1. According to the lecture on project management standards, what does PMBOK stand for?

A) Project Management Baseline of Knowledge
B) Process Management Body of Knowledge
C) Project Management Body of Knowledge
D) Program Management Baseline of Knowledge

Source: A8-PtgugnGE | Difficulty: Easy

Q2. In Scrum, what is the term for a fixed-duration iteration, typically one month long, during which a potentially shippable product increment is developed?

A) Cycle
B) Sprint
C) Phase
D) Increment

Source: 0JYkgfgKn8I | Difficulty: Easy

Q3. What is the primary purpose of a Work Breakdown Structure (WBS) in project scheduling?

A) To calculate the project's total cost
B) To assign resources to team members
C) To create a marketing plan for the product
D) To hierarchically decompose project work into smaller, manageable components

Source: AvwZ7JO8LNk | Difficulty: Easy

Q4. In cost-benefit analysis, what does ROI stand for?

A) Rate of Interest
B) Return on Investment
C) Risk of Investment
D) Revenue of Initiative

Source: 8UlgpszPK6w | Difficulty: Easy

Q5. What is the primary characteristic of a 'Functional Organization' structure?

A) Teams are formed for a single project and then disbanded
B) Employees report to multiple project managers simultaneously
C) Employees are grouped by specialized functions like design or testing
D) All decisions are made by consensus among team members

Source: MpNjo7fbm3M | Difficulty: Easy

Q6. What is the main goal of the 'Cleanroom Technique' in software development?

A) To test software in a sterile hardware environment
B) To prevent defects from entering the code in the first place
C) To write code that is easy for new developers to clean up
D) To ensure all developers have a clean and organized workspace

Source: 4ZLbpoBcdo8 | Difficulty: Easy

Q7. What is the primary purpose of a feasibility study in the software development lifecycle?

A) To write the initial code for the project
B) To determine if a project is viable before committing significant resources
C) To train the new development team
D) To create the final user manual

Source: yC0A6_nYPYQ | Difficulty: Easy

Q8. According to the lecture, what is the most significant drawback of the 'Price to Win' estimation method?

A) It is too time-consuming to calculate
B) It requires extensive historical data
C) The estimated costs do not accurately reflect the actual work required
D) It can only be used for small projects

Source: wqyJF-Gbdp0 | Difficulty: Easy

Q9. What is the primary purpose of a Project Closure Report?

A) To request additional funding for the project
B) To assign new tasks to the project team
C) To document project outcomes, performance, and lessons learned for future projects
D) To market the newly completed software product

Source: TfvonvpVeTE | Difficulty: Easy

Q10. In PERT/CPM, what is the sequence of activities with zero slack that determines the minimum project duration called?

A) The Subcritical Path
B) The Optimal Route
C) The Critical Path
D) The Baseline Schedule

Source: gqtKDlYSJc | Difficulty: Easy

Q11. In a PERT/CPM forward pass, how is the Earliest Start (ES) time determined for an activity with multiple predecessors?

A) It is the earliest (minimum) of the Earliest Finish (EF) times of all its predecessors.
B) It is the average of the Earliest Finish (EF) times of all its predecessors.
C) It is the latest (maximum) of the Earliest Finish (EF) times of all its immediate predecessors.
D) It is determined by the project manager's discretion.

Source: 8w2gm0AO-KY | Difficulty: Medium

Q12. According to Herzberg's Two-Factor Theory, which of the following is considered a 'Motivator' rather than a 'Hygiene Factor'?

A) Company policy
B) Pay
C) Working conditions
D) Recognition of effort and performance

Source: TQWJYqOzs0g | Difficulty: Medium

Q13. What is the key difference between the RAD model and the traditional prototyping model?

A) The RAD model does not involve customer feedback.
B) In RAD, the prototype is refined into the final product, whereas in the prototyping model, it is typically thrown away.
C) The RAD model is only suitable for large, multi-year projects.
D) The prototyping model uses a 'time box' approach, while RAD does not.

Source: 0O6fJSkafOo | Difficulty: Medium

Q14. In Earned Value Analysis, if a project has a Schedule Performance Index (SPI) of 0.88 and a Cost Performance Index (CPI) of 0.64, what does this indicate?

A) The project is ahead of schedule and under budget.
B) The project is ahead of schedule but over budget.
C) The project is behind schedule and over budget.
D) The project is behind schedule but under budget.

Source: I81fOjyzb0E | Difficulty: Medium

Q15. What is the primary reason software reliability measurement is more complex than hardware reliability measurement?

A) Software is more expensive than hardware.
B) Hardware failures are always caused by user error.
C) Software failures are due to design flaws, not physical wear, and reliability changes with each bug fix.
D) There are no standardized metrics for hardware reliability.

Source: jUUcunXtii0 | Difficulty: Medium

Q16. How does a 'Project Organization' structure differ from a 'Functional Organization'?

A) A Project Organization is permanent, while a Functional Organization is temporary.
B) In a Project Organization, employees report to a functional manager, while in a Functional Organization, they report to a project manager.
C) A Project Organization offers less job variety for team members.
D) In a Project Organization, employees are dedicated to a single project team for its duration, whereas in a Functional Organization, they are drawn from specialized departments.

Source: MpNjo7fbm3M | Difficulty: Medium

Q17. In the context of contract management, what is the main purpose of a Memoranda of Agreement (MoA)?

A) To serve as the final legal contract for payment.
B) To document the minutes of all project meetings.
C) To freeze the technical specifications of a solution so subsequent tenders can be judged primarily on price.
D) To outline the marketing strategy for the new product.

Source: -hlbuCXjNGY | Difficulty: Medium

Q18. According to the lecture, what is a primary limitation of the classical waterfall model that the iterative waterfall model attempts to address?

A) Its lack of clear phases.
B) Its inability to handle rework and defects discovered in later phases.
C) Its unsuitability for projects with stable requirements.
D) Its over-reliance on customer feedback.

Source: J1-LpZr9GSQ | Difficulty: Medium

Q19. What is the key difference between a 'slip chart' and a 'Gantt chart' for project monitoring?

A) A Gantt chart is used for cost monitoring, while a slip chart is for schedule monitoring.
B) A slip chart visually emphasizes deviations from the schedule, while a Gantt chart provides a static snapshot of planned durations.
C) Only Gantt charts can show the critical path.
D) Slip charts are used for initial planning, while Gantt charts are for post-implementation review.

Source: bvt6NAabXPA | Difficulty: Medium

Q20. In the context of risk management, what does 'Risk Reduction Leverage (RRL)' measure?

A) The total number of risks in a project.
B) The probability of a risk occurring.
C) The cost-effectiveness of a risk handling technique.
D) The amount of time a project is delayed by a risk.
Source: PehmL5jOmW8 | Difficulty: Medium

Q21. What is the primary purpose of the 'Value Adjustment Factor (VAF)' in Albrecht/IFPUG function point analysis?

A) To convert function points into lines of code.
B) To calculate the initial raw count of functional units.
C) To determine the number of user inputs and outputs.
D) To refine the raw functional size by accounting for technical and environmental complexity.
Source: MzhN6eZIUNU | Difficulty: Medium

Q22. In the SEI CMM, what distinguishes Level 4 (Managed) from Level 3 (Defined)?

A) Level 4 introduces basic project management, while Level 3 is chaotic.
B) Level 4 focuses on continuous process improvement, while Level 3 focuses on repeating success.
C) Level 4 introduces quantitative measurement and control of processes, while Level 3 focuses on standardizing and documenting them.
D) Level 4 requires peer reviews, while Level 3 does not.
Source: k6I04YmQWl8 | Difficulty: Medium

Q23. According to Putnam's model, if a project's development time is compressed by 50% (e.g., from 2 years to 1 year), by what factor does the required effort increase?

A) 2 times
B) 4 times
C) 8 times
D) 16 times
Source: _KdYrzzMuBo | Difficulty: Hard

Q24. In a PERT/CPM backward pass, how is the Latest Finish (LF) time determined for an activity that is a 'burst point' (i.e., has multiple successors)?

A) It is the latest (maximum) of the Latest Start (LS) times of all its successors.
B) It is the smallest (minimum) of the Latest Start (LS) times of all its immediate successors.
C) It is the average of the Latest Start (LS) times of all its successors.
D) It is equal to the project's overall earliest finish time.
Source: 8w2gm0AO-KY | Difficulty: Hard

Q25. In Halstead's Software Science, how is Programming Effort (E) calculated, and what does it represent?

A) E = V * L, representing the total lines of code.
B) E = N * log2(η), representing the program volume.
C) E = V^2 / V*, representing the number of mental discriminations required to implement the program.
D) E = V / S, representing the total development time in months.
Source: w_Ka3XuXEbw | Difficulty: Hard

Q26. What is a fundamental difference in how the Jelinski Moranda model and the Littlewood and Verall’s model account for reliability growth?

A) Jelinski Moranda is a step function model, while Littlewood and Verall's is linear.
B) Littlewood and Verall’s model accounts for negative reliability growth and diminishing returns, while Jelinski Moranda assumes reliability improvement is proportional to remaining faults.
C) Jelinski Moranda is used for hardware reliability, while Littlewood and Verall's is for software.
D) Littlewood and Verall’s model assumes each bug fix contributes equally to reliability, which Jelinski Moranda refutes.

Source: M82c48MiL4w | Difficulty: Hard

Q27. A project manager is using the 0/100 technique for earned value. A task with a Planned Value (PV) of $10,000 is scheduled for completion by week 4. At the end of week 4, the task is 90% complete with an Actual Cost (AC) of $8,000. What are the Earned Value (EV), Cost Variance (CV), and Schedule Variance (SV)?

A) EV = $9,000, CV = $1,000, SV = -$1,000
B) EV = $0, CV = -$8,000, SV = -$10,000
C) EV = $10,000, CV = $2,000, SV = $0
D) EV = $0, CV = $2,000, SV = -$10,000

Source: 88pXUUtmnbg | Difficulty: Hard

Q28. Why is an organization with ISO 9001 certification prohibited from advertising its individual products as 'ISO 9000 compliant'?

A) Because ISO 9001 certification is only valid for one year.
B) Because the ISO organization charges a separate fee for product-level advertising.
C) Because ISO 9001 certifies the organization's quality process, not the individual products themselves.
D) Because only organizations in the manufacturing sector can advertise product compliance.

Source: ODQNex8xtjU | Difficulty: Hard

Q29. In the context of team structures, why is a hierarchical team structure considered more suitable for large projects than a democratic team structure?

A) A hierarchical team is less expensive to manage.
B) A hierarchical structure scales effectively by distributing leadership, whereas the N-squared communication paths in a democratic team become highly inefficient as team size increases.
C) Democratic teams are only suitable for maintenance projects.
D) Hierarchical teams do not require a project manager.

Source: wjd0YMixlJA | Difficulty: Hard

Q30. In COCOMO II's Application Composition Model, what three components are counted to determine the Number of Application Points (NAP)?

A) Number of lines of code, number of developers, and project duration.
B) Number of use cases, number of actors, and number of classes.
C) Number of separate screens, number of reports produced, and number of modules.
D) Number of hardware devices, number of databases, and number of network connections.

Source: JXiFgnkKGAQ | Difficulty: Hard

Answer Key

1: C	2: B	3: D	4: B	5: C
6: B	7: B	8: C	9: C	10: C
11: C	12: D	13: B	14: C	15: C
16: D	17: C	18: B	19: B	20: C
21: D	22: C	23: D	24: B	25: C
26: B	27: B	28: C	29: B	30: C